

Project Phantom

Ethical, Security, and Governance Challenges of Deepfake Technology in the Era of Generative Artificial Intelligence

Walther Del Orbe, Nawaz Syed, Kwaku Dei Amoani, Gwendolyn Vongkasemsiri

The Beacom College of Computer and Cyber Sciences

Dakota State University, Madison, SD, USA

Abstract

Deepfake technologies enabled by generative artificial intelligence are rapidly transforming digital media creation and manipulation. These technologies leverage Generative Adversarial Networks (GANs), diffusion models, and transformer-based architectures to produce highly realistic synthetic audio, video, and images that convincingly mimic authentic content. While these tools offer measurable benefits in education, accessibility, and creative industries, they simultaneously introduce substantial risks — including large-scale misinformation, identity impersonation, financial fraud, and the erosion of public trust in digital communication. This research examines deepfake technologies through three interconnected lenses: ethical implications, cybersecurity threats, and governance challenges. Through a structured literature review spanning 2021–2026 and a comparative analysis of detection techniques and policy frameworks across jurisdictions, we identify critical gaps in existing mitigation approaches. Our findings demonstrate that: (1) detection methods are consistently outpaced by advances in generative modeling; (2) legal frameworks remain fragmented and jurisdiction-specific; and (3) effective governance requires integrated technical, ethical, and policy responses. This paper contributes a synthesized multi-dimensional analysis intended to guide responsible development, deployment, and regulation of generative AI systems.

synthetic outputs, commonly termed deepfakes, encompass manipulated images, videos, and audio that replicate real individuals' faces, voices, and behavioral characteristics with striking verisimilitude.

The proliferation of deepfake tools presents a dual-use challenge. On one hand, these capabilities drive innovation in film production, virtual education, medical simulation, and accessibility tools for individuals with disabilities. On the other, they are exploited for political disinformation, financial fraud, non-consensual intimate imagery, and large-scale identity impersonation. Open-source frameworks such as Stable Diffusion, StyleGAN variants, and numerous face-swap applications have reduced the technical barrier for malicious actors to near zero, decoupling harmful capability from technical expertise and creating an unprecedented asymmetry between the cost of creating a convincing deepfake and the severity of its societal consequences.

Digital trust — underpinning modern communication, financial systems, and democratic processes — is increasingly threatened by synthetic media. Traditional authenticity verification mechanisms such as visual inspection, metadata review, and trusted-source evaluation are becoming insufficient against AI-generated content. The “liar’s dividend” further complicates matters: as deepfakes proliferate, even genuine recordings may be dismissed as fabrications, eroding confidence in authentic evidence and undermining accountability in legal, journalistic, and political contexts.

1 Introduction

Recent advances in artificial intelligence have fundamentally altered the landscape of digital media creation. Generative models — including Generative Adversarial Networks (GANs), diffusion models, and transformer-based architectures — now enable the production of synthetic media nearly indistinguishable from authentic recordings. These

2 Related Work

The literature on deepfake technologies spans computer vision, natural language processing, cybersecurity, ethics, and law. This section reviews the most relevant recent contributions across four subdomains, explicitly compares their methodological approaches and findings, and identifies the research

gap that this paper addresses. Prior surveys have typically examined one dimension in depth — detection methods, societal impact, or governance — but have not integrated all three into a unified comparative analysis. This paper’s primary contribution is precisely that integration.

2.1 Deepfake Detection Methods

Detection research is the most active sub-field in the deepfake literature, yet it is also characterized by a fundamental and persistent tension: benchmark accuracy and real-world generalization point in opposite directions. Khan et al. (2025) provide the most comprehensive adversarial robustness review, categorizing detection methods into visual artifact analysis, biological signal detection, and neural network-based classifiers. Critically, they demonstrate that no single method achieves robust performance across all modalities — a cross-method comparative finding that contrasts with the single-method optimism typical of individual detection papers. The present study adopts Khan et al.’s categorization and extends it with two additional evaluation dimensions absent from their framework: real-time deployment feasibility and adversarial robustness under distribution shift.

Three papers directly address the generalization crisis, but from different angles. Soundarya and Gururaj (2026) approach it architecturally, showing that detectors fail when generative architecture changes (GAN to diffusion). Gong and Li (2024) approach it statistically, demonstrating that benchmark datasets are constructed under conditions that inflate accuracy and do not reflect real-world distribution. Lin et al. (2025) approach it operationally, documenting consistent degradation when detectors are deployed outside their training environment. Together these three studies converge on the same conclusion — detection accuracy claims cannot be taken at face value — but each reaches that conclusion through a distinct methodology, which explains why the problem has persisted despite being identified by multiple groups. Liu et al. (2025) complete the picture by establishing that diffusion-based generation is qualitatively different from GAN-based generation in terms of artifact production, making GAN-trained detectors not merely weaker on diffusion content but architecturally mismatched. The one clear advance is Doloriel et al. (2025), whose frequency-domain masking achieves the highest cross-architecture generalization of any method in the reviewed literature

— the only result that addresses the generalization crisis rather than merely documenting it.

2.2 Societal and Psychological Impact

Farid (2025) argues that deepfake technologies pose a systemic threat to public trust in journalism and democratic institutions, and that technological countermeasures alone are insufficient without complementary policy frameworks. This paper directly incorporates this argument into Section 6 by advocating for multi-stakeholder accountability structures rather than purely technical solutions. Clark and Lewandowsky (2026) demonstrate the "continued influence effect" — AI-generated deepfake videos continue to shape viewer beliefs even when explicit authenticity warnings are provided — challenging assumptions that disclosure alone can mitigate persuasive impact. Their findings are connected in Section 8 of this paper to the ethical analysis of the liar’s dividend.

2.3 Linguistic and Multimodal Detection

Martinek (2025) and Singh et al. (2025) represent two complementary advances in detection, but they operate at different levels of abstraction. Martinek demonstrates that linguistic signals alone — without access to any audio or video — can discriminate deepfake advertisements from authentic ones with meaningful accuracy. Singh et al. establish that combining linguistic signals with visual and audio modalities further improves performance. The practical implication is a detection hierarchy: linguistic analysis is cheap and architecture-agnostic; audio analysis adds coverage at moderate cost; visual analysis adds further coverage at higher cost; and ensemble integration yields maximum performance at maximum cost. This hierarchy is not made explicit in either paper but is the appropriate synthesis for practitioners making deployment tradeoffs. The key gap neither paper addresses is how to select an optimal point on this hierarchy for a given deployment context — a question this paper’s five-dimension framework in Table 2 is designed to help answer.

2.4 Governance and Legal Frameworks

Bateman et al. (2025), writing in Computer Law and Security Review, propose a legal framework for deepfake governance grounded in human rights principles, centering the right to informational self-determination and the right to truth. This paper builds on that framework by evaluating it against

concrete jurisdictional examples, including the EU AI Act, U.S. state legislation, and Chinese national regulations, revealing cross-border inconsistencies not yet comprehensively synthesized in prior work. Patel et al. (2026) integrate detection and content provenance tracking into an AI-driven multi-stakeholder governance model that includes platforms, regulators, and civil society; this paper adopts that model as the basis for governance recommendations in Section 7.3 while identifying practical implementation barriers the original framework does not fully address.

2.5 Research Gap and Contribution of This Paper

Across the four sub-domains reviewed above, a consistent structural gap emerges: existing work examines each dimension of the deepfake problem — detection, societal impact, linguistic analysis, and governance — in relative isolation. Detection surveys do not account for governance constraints on detection deployment. Governance proposals do not engage seriously with the technical limitations of detection that would make enforcement possible. Societal impact research identifies the scale of harm without connecting it to specific governance or detection interventions that could reduce it. No prior work synthesizes all three dimensions — technical detection, cybersecurity risk, and governance — into a single integrated analysis that allows the constraints and insights of each to inform the others.

This gap has a concrete practical consequence. A policymaker reading Khan et al. (2025) learns what detection methods exist but not which ones could actually be mandated at platform scale. A platform engineer reading Bateman et al. (2025) learns what rights framework should govern synthetic media but not what technical standards would implement those rights. A researcher reading Lin et al. (2025) learns how detection fails in the real world but not what governance levers could compensate for that failure. The interdependency among these three dimensions means that gaps in any one cannot be addressed without understanding the others.

The contribution of this paper is to provide precisely that integrated synthesis. It does not propose a new detection algorithm, a new legal framework, or a new empirical study of deepfake harm. Instead, it maps the relationships among existing knowledge across all three dimensions, identifies where the constraints of one dimension invalidate assump-

tions in another, and derives recommendations that are simultaneously technically feasible, ethically grounded, and governmentally actionable. This is the research gap this paper addresses.

3 Research Questions

This study is guided by two primary research questions that define the analytical scope of the paper. Each question is addressed through the structured literature review and comparative framework described in Section 4, and each is answered directly in Section 10 (Discussion and Findings).

RQ1. What ethical challenges arise from the development and use of deepfake technologies, particularly regarding consent, identity protection, and human dignity?

RQ2. What cybersecurity and fraud risks are introduced by deepfake technologies, and how effective are current detection approaches?

These two questions capture the core tension at the heart of the deepfake problem: harm to individuals and institutions on one side, and the adequacy of technical and governance countermeasures on the other. Together they structure the analysis in Sections 5 through 8 and are revisited with direct answers in the conclusion.

4 Methods

This study employs a structured literature review methodology combined with a comparative analytical framework. The review focuses on peer-reviewed publications, conference proceedings, and technical reports published from 2021–2026 addressing deepfake detection, ethics of generative AI, and policy responses to synthetic media misuse.

4.1 Literature Search and Selection

Literature was identified through systematic searches of four databases: Google Scholar, IEEE Xplore, ACM Digital Library, and arXiv. Search terms included “deepfake detection,” “synthetic media ethics,” “generative AI governance,” “GAN deepfake,” “diffusion model detection,” and “deepfake policy,” applied individually and in Boolean combinations. The search was restricted to publications from January 2021 through April 2026 to ensure relevance to current generative AI capabilities.

The screening process proceeded in four stages. The initial database search returned approximately 340 candidate titles. Title and abstract screening

against inclusion and exclusion criteria reduced this to 60 sources. Full-text review for methodological quality and dimensional relevance yielded 30 candidates. Final selection, prioritizing recency, empirical grounding, and cross-dimensional coverage, produced 15 primary sources. A cross-reference check of cited works in the selected sources identified no additional materials meeting all inclusion criteria. The resulting corpus is summarized in Table 5.

Inclusion criteria required that a source: (1) be published in a peer-reviewed venue, conference proceeding, or reputable preprint archive; (2) address at least one of the three analytical dimensions; (3) fall within the 2021–2026 date range; and (4) be available in English. Sources were excluded if they focused solely on generative model development without addressing detection, risk, or governance, or if they lacked verifiable peer review.

T = Technical Detection C = Cybersecurity Risk G = Governance & Ethics

4.2 Analytical Framework

The analytical framework is organized across three dimensions — technical detection, cybersecurity risk, and governance — and this structure requires justification, as it is a substantive choice rather than a neutral container. Technical detection is included because it represents the primary operational mechanism by which individuals and platforms can identify synthetic media at the point of encounter; without understanding what detection can and cannot do, governance recommendations are untethered from implementation reality. Cybersecurity risk is included because deepfake threats have evolved beyond disinformation into direct attacks on financial systems and authentication infrastructure. Governance is included because technical detection alone cannot address deepfake harm.

The three dimensions are mutually constraining: the feasibility of governance depends on the state of detection, and the priorities of detection research should be shaped by governance mandates.

Detection approaches are compared across five criteria:

- Benchmark accuracy
- Generalization to unseen deepfake types
- Adversarial robustness
- Computational cost

- Real-time deployment feasibility

These criteria capture both academic performance and real-world deployment constraints.

4.3 Project Repository

All supplementary materials for this project are maintained in a public GitHub repository hosted under the DSU Cyberops organization. The repository is organized into clearly labeled folders corresponding to each section of this paper and is designed to support reproducibility and further research.

Repository: <https://github.com/DSUcyberops/csc727>

The repository includes:

- Annotated bibliography
- PRISMA-style search documentation
- Detection comparison data
- Governance analysis
- Final paper

5 Cybersecurity and Fraud Risks

Deepfake technologies introduce a broad and evolving spectrum of cybersecurity and fraud risks that extend far beyond disinformation into financial systems, authentication infrastructure, and legal processes. The following subsections document primary attack categories, drawing on documented incidents and the threat typology presented in Table 1.

5.1 Voice Cloning and Business Email Compromise

Voice cloning represents one of the most immediately impactful deepfake threats. Attackers generate synthetic audio mimicking an executive or trusted colleague to request emergency financial transfers or sensitive credentials. This attack vector has been operationalized alongside business email compromise (BEC) campaigns, where synthetic voice calls reinforce fraudulent email requests, overriding verification protocols. Documented incidents have demonstrated multi-million-dollar losses attributed to voice cloning attacks against European and Asian financial institutions. The convergence of voice cloning with social engineering is particularly dangerous because it circumvents authentication mechanisms that rely on voice recognition or human judgment, the two factors most likely to catch traditional BEC attempts.

Table 1: Summary of Reviewed Sources by Author, Year, Venue, and Analytical Dimension

#	Author(s)	Year	Venue / Type	T	C	G
1	Clark & Lewandowsky	2026	Communications Psychology (Journal)	✓	✓	
2	Doloriel et al.	2025	arXiv preprint (CS)	✓		
3	Farid	2025	PNAS Nexus (Journal)	✓	✓	
4	Gong & Li	2024	Electronics (Journal)	✓		
5	Guarnera et al.	2023	arXiv preprint (CS)	✓		
6	Khan et al.	2025	arXiv preprint (CS)	✓		
7	Bateman et al.	2025	Computer Law & Security Review			✓
8	Lin et al.	2025	arXiv preprint (CS)	✓	✓	
9	Liu et al.	2025	Intl. Journal of Intelligent Systems	✓		
10	Martinek	2025	COLING 2025 (Conference)	✓		
11	Patel et al.	2026	Digital Health & Media (Journal)	✓	✓	
12	Singh et al.	2025	Journal of AI Research	✓		
13	Soundarya & Gururaj	2026	Discover Applied Sciences	✓		
14	Soundarya et al.	2025	Discover Computing (Journal)	✓		
15	European Parliament	2024	Legislative text (EU AI Act)			✓

T = Technical Detection C = Cybersecurity Risk G = Governance & Ethics

5.2 Identity Impersonation in Authentication Systems

Deepfake video enables impersonation in remote verification contexts, including video-based know-your-customer (KYC) processes used by financial institutions and government identity verification systems. Many organizations adopted video verification as a security improvement without accounting for synthetic video injection. Real-time face-swap technologies further exacerbate this risk by enabling live impersonation during video calls, directly targeting multi-factor authentication and liveness detection systems. As biometric authentication becomes more prevalent across financial, governmental, and enterprise contexts, the risk surface introduced by deepfake impersonation grows correspondingly larger.

5.3 Misinformation and Political Manipulation

Synthetic media is increasingly deployed in coordinated information operations designed to manipulate public opinion, discredit public figures, or fabricate crisis scenarios. The 2024 and 2025 electoral cycles in multiple countries documented use of AI-generated content in political advertising and disinformation campaigns, raising urgent concerns about the integrity of digital political discourse and the stability of democratic institutions. Deepfake-driven political manipulation is particularly difficult to counter because viral spread through social networks typically outpaces verification and correction, and because the psychological impact persists even after corrections are issued.

5.4 Extended Attack Surface

Beyond the primary categories above, deepfakes enable a range of additional cybersecurity threats that are likely to grow in severity as generation becomes more accessible:

- Biometric authentication bypass using face or voice spoofing against FIDO2 and liveness detection systems
- Insider impersonation attacks exploiting organizational trust relationships and privileged access pathways
- Synthetic evidence generation to fabricate or undermine the integrity of legal proceedings
- Market manipulation using fabricated executive or regulatory statements distributed through authentic-appearing channels
- Long-term social engineering campaigns building false rapport through sustained video deepfake personas
- Fraudulent customer support impersonation targeting vulnerable or elderly populations

Four attack types are rated Critical due to direct threats to financial systems, legal systems, or democratic institutions. The pattern across Critical attacks is consistent: each exploits an institutional trust relationship at scale by weaponizing the assumption that media artifacts are authentic — an assumption deepfake technology systematically invalidates.

6 Deepfake Detection Techniques

Detection methods have advanced in parallel with generative capabilities, yet remain in a persistent adversarial deficit. As generative architectures evolve — most significantly in the shift from GANs to diffusion models — detection approaches trained on prior-generation content degrade rapidly. This section categorizes and evaluates current detection techniques across five performance dimensions: benchmark accuracy, generalization, adversarial robustness, computational cost, and real-time deployment feasibility.

6.1 Visual Artifact Detection

These approaches identify inconsistencies in lighting, facial geometry, blending boundaries, and rendering artifacts at the pixel level. Effective against

early GAN-generated media, they are substantially weakened by diffusion-based models that produce fewer compression and blending artifacts. Benchmark accuracy on GAN test sets exceeds 90%, but generalization to out-of-distribution and cross-architecture content drops to near-chance levels, making this approach increasingly unreliable as the field's primary generation paradigm shifts.

6.2 Biological Signal Analysis

These techniques analyze physiological signals — eye blink rates, micro-expression timing, subtle breathing motion, and pulse estimation from skin color variation — that are difficult to synthesize authentically. Effective against first- and second-generation deepfakes, newer models trained on high-frame-rate video datasets now simulate biological patterns with sufficient fidelity to evade detection, particularly when high-quality source video is available to the attacker.

6.3 Audio Spectral Analysis

Audio-based detection analyzes spectral anomalies in synthetic speech, including inconsistencies in pitch contours, formant trajectories, and waveform characteristics. Advanced voice cloning architectures (VITS, VALL-E) have substantially narrowed the detectability gap, particularly for short audio clips under three seconds where spectral context is limited and where most real-world BEC voice clips fall.

6.4 Metadata and Provenance Analysis

Metadata examination inspects file origin, compression history, and encoding inconsistencies. Effective when metadata is present and unmodified, this approach fails when metadata is stripped — a trivial operation for a technically sophisticated attacker. Content credential standards (C2PA) aim to address this gap through cryptographic provenance chains attached at point of creation, but adoption remains limited and does not retroactively address existing synthetic content.

6.5 Machine Learning Classifiers and Ensembles

Neural network-based classifiers achieve state-of-the-art benchmark performance but require continuous retraining as generative models evolve — creating a maintenance burden that limits practical longevity without dedicated infrastructure. Multimodal ensemble systems consistently outperform

Table 2: Deepfake Attack Typology and Organizational Risk Assessment

Attack Type	Primary Target	Impact Domain	Detection Difficulty	Risk Level
Voice Cloning (BEC)	Financial institutions	Financial fraud	High	Critical
Video KYC Impersonation	Banks, government services	Identity theft	Very High	Critical
Political Disinformation	Public, media institutions	Democratic integrity	High	Critical
Biometric Spoofing	Authentication systems	Unauthorized access	High	High
Non-consensual Media	Private individuals	Psychological/legal harm	Moderate	High
Synthetic Legal Evidence	Courts, law enforcement	Obstruction of justice	Very High	Critical
Market Manipulation	Financial markets	Economic disruption	High	High
Social Engineering Video	Individuals, organizations	Credential theft	Moderate	Moderate

single-modality approaches but impose substantial computational costs, typically making real-time deployment infeasible at scale. Frequency-domain masking (Doloriel et al., 2025) offers the best generalization-to-cost tradeoff of any approach in the reviewed literature, achieving high cross-architecture generalization at moderate computational expense.

Frequency-domain masking achieves the best generalization score of any single approach and is the only method rated High across generalization, adversarial robustness, and benchmark accuracy simultaneously. Multimodal ensembles remain the overall performance leader but are infeasible for real-time deployment without dedicated infrastructure. Visual artifact detection, despite high benchmark accuracy, cannot be relied upon as generative architectures continue their shift toward diffusion-based generation.

Note: Values in Table 4 are approximations derived from ranges reported across cited literature benchmarks. They are intended to illustrate relative performance differences across methods, not to assert precise empirical accuracy.

7 Governance and Mitigation Approaches

Governance frameworks for deepfake technologies remain nascent and fragmented. Regulatory responses have emerged unevenly across jurisdictions, creating enforcement gaps that allow synthetic media misuse to proliferate transnationally. This section reviews legislative approaches, technical mitigation strategies, and multi-stakeholder governance models, and identifies critical gaps in the current landscape.

7.1 Legislative Landscape

In the United States, California and Texas have enacted legislation targeting non-consensual deepfake pornography and political deepfakes; federal legislation has been proposed — including the DEFIANCE Act — but not enacted as of this writing. The absence of federal legislation creates inconsistent protection for individuals depending on their state of residence and leaves cross-state cases without a clear legal framework. The EU AI Act takes a risk-based regulatory approach, requiring high-risk AI applications to disclose synthetic content generation and mandating watermarking of AI-generated media; its risk-tiering system represents a more systematic approach than U.S. fragmentation, but faces enforcement challenges across member states. China has implemented comprehensive deepfake regulations requiring content labeling and platform accountability — among the most assertive national frameworks globally, though critics note enforcement has been applied inconsistently and primarily in politically sensitive contexts.

A common gap persists across all jurisdictions: none of the existing frameworks adequately addresses the transnational character of synthetic media production and distribution. A deepfake generated in one jurisdiction, distributed via a platform headquartered in a second, and consumed in a third may fall through all applicable regulatory frameworks simultaneously.

7.2 Technical Mitigation Strategies

Technical approaches complement legislative efforts across the content lifecycle:

- AI watermarking — embedding impercepti-

Table 3: Comparative Analysis of Deepfake Detection Approaches

Detection Method	Benchmark Accuracy	Generalization	Adversarial Robustness	Compute Cost	Real-time Feasible
Visual Artifact	High (GAN)	Low (diffusion)	Low	Low	Yes
Biological Signal	Moderate	Moderate	Moderate	Low	Yes
Audio Spectral	Moderate-High	Moderate	Low-Moderate	Low	Yes
Metadata Analysis	Moderate	Low	Very Low	Very Low	Yes
ML Classifier	High	Moderate	Moderate	Moderate	Partial
Multimodal Ensemble	Highest	Moderate-High	Moderate-High	High	No
Freq-Domain Masking	High	High	High	Moderate	Partial

Table 4: Detection Accuracy vs. Cross-Architecture Generalization (Approximate)

Detection Method	Benchmark Accuracy	Generalization Score
Visual Artifact	~92%	~38%
Biological Signal	~78%	~62%
Audio Spectral	~84%	~58%
Metadata Analysis	~71%	~32%
ML Classifier	~91%	~66%
Multimodal Ensemble	~96%	~74%
Freq-Domain Masking	~88%	~81%

ble provenance signals at generation time to enable attribution

- C2PA content credentials — cryptographic signing recording a verifiable chain of custody from creation to publication
- Detection APIs — platform-integrated tools flagging potentially synthetic content for human review before amplification
- Platform moderation policies combining automated and human review with community standards enforcement
- Enhanced liveness detection in biometric authentication systems to resist face and voice spoofing

Each approach carries limitations: watermarks can be removed by post-processing; detection systems require continuous retraining as generative models evolve; moderation at scale introduces accuracy-throughput tradeoffs; and liveness detection systems face the same adversarial dynamic as other detection approaches. Technical mitigation alone is therefore insufficient and must be integrated with the legal and governance frameworks described in Sections 7.1 and 7.3.

7.3 Multi-Stakeholder Governance Model

Effective governance requires coordination across developers, platforms, regulators, researchers, and civil society. Developers must implement responsible disclosure and watermarking by design. Platforms must enforce content moderation and provide detection infrastructure integrated with distribution pipelines. Regulators must establish liability frameworks, mandate transparency requirements, and fund independent verification bodies. Researchers must advance detection capabilities and publish findings in ways that support defensive applications. Civil society must advocate for affected individuals, audit platform enforcement for bias and consistency, and ensure governance frameworks reflect the interests of marginalized populations. No single actor can address the challenge unilaterally.

The primary implementation challenge is coordination. Without shared technical standards, harmonized legal definitions, and binding international agreements, responsibilities diffuse to the point where no actor is meaningfully accountable. This paper recommends that international bodies — including the United Nations and the OECD — develop binding minimum standards for synthetic media transparency, and that liability frameworks explicitly assign responsibility to the point in the

Table 5: Jurisdictional Governance Frameworks for Deepfake Technologies

Jurisdiction	Primary Legislation	Key Requirements	Enforcement Body	Maturity
United States (Federal)	No federal law enacted	Proposed DEFIANCE Act; state-level varies	FTC / State AGs	Low
United States (CA/TX)	AB 602, AB 730; TX HB 4337	Non-consensual deepfake prohibition; political disclosure	State Attorneys General	Moderate
European Union	EU AI Act (2024)	Watermarking required; transparency obligations; platform duties	EU AI Office	High
China	Deepfake Regulations (2022, 2023)	Mandatory labeling; platform liability; real-name registration	CAC	High
United Kingdom	Online Safety Act (2023)	Harmful content moderation; CSAM deepfake provisions	Ofcom	Moderate

production and distribution chain where prevention was most feasible.

8 Ethics Statement

Deepfake technologies raise fundamental ethical concerns that transcend technical and legal dimensions. This section addresses five interconnected issues: consent and identity ownership, epistemic harm, fairness and bias in detection systems, accountability and transparency, and the ethical conduct of this research itself.

8.1 Consent and Identity Ownership

The creation of synthetic media depicting a real individual without consent constitutes a profound violation of personal autonomy and dignity. Non-consensual deepfakes — including fabricated intimate imagery, false statements attributed to public figures, and manipulated recordings of private individuals — cause severe and lasting psychological, reputational, and economic harm. Research on non-consensual intimate deepfakes documents outcomes including acute anxiety, depression, social withdrawal, and reputational damage that may be impossible to fully remediate once synthetic content has been distributed. Ethical frameworks governing deepfake development must centralize individuals’ rights to control their digital likeness, voice, and identity representation. The principle of consent must also extend to training data: current industry practices frequently rely on scraped datasets collected without explicit consent, raising ethical concerns even when the resulting model is used for ostensibly benign purposes.

8.2 The Liar’s Dividend and Epistemic Harm

As deepfakes proliferate, individuals may invoke the possibility of synthetic fabrication to deny authentic evidence — the “liar’s dividend.” This epistemic harm undermines accountability mechanisms across legal, political, and social contexts. Documented cases already exist in which defendants in legal proceedings have contested the authenticity of video evidence by alleging deepfake manipulation, regardless of whether such manipulation occurred. Mitigating this effect requires authenticated content provenance standards, media literacy programs, and legal standards for the admissibility and authentication of digital evidence alongside technical detection tools.

8.3 Fairness and Bias in Detection Systems

Detection systems introduce their own ethical concerns. Studies consistently identify that detection models perform inconsistently across racial, gender, and age groups — historically marginalized populations experience both higher false positive rates (incorrectly flagged as synthetic when genuine) and in some cases lower true positive rates (missed actual deepfakes). These disparities reflect training data biases and must be addressed through diverse dataset curation, algorithmic bias auditing, and equitable evaluation standards mandated as prerequisites for deployment. A detection system that systematically disadvantages already vulnerable populations compounds existing inequities, and its uncritical deployment at scale may cause harm disproportionate to any security benefit.

8.4 Accountability and Transparency

The distributed and often anonymous nature of synthetic media generation complicates accountability. Assigning responsibility across a chain of actors — model developers, hosting platforms, content distributors, and end users — presents significant legal and ethical challenges. Transparency requirements, including mandatory disclosure of AI-generated content and retention of generation provenance logs, are essential preconditions for meaningful accountability frameworks and legal redress. Developers should be required to log generation events in a manner that supports forensic attribution; platforms should be required to disclose when AI-generated content is present; and regulators should be empowered to mandate and audit compliance with these requirements.

8.5 Research Ethics

This research relies exclusively on published academic sources and does not involve the creation, analysis, or distribution of actual deepfake content. No human subjects were involved, and no proprietary or sensitive data were accessed. All findings are intended to support defensive and protective applications. The authors acknowledge that research on synthetic media must be conducted carefully to avoid inadvertently advancing harmful capabilities, and that publication of detection limitations must be balanced against the risk of informing adversarial evasion.

9 Limitations

This study carries several limitations that should be considered when interpreting findings. First, the research relies exclusively on published literature without original experimental data or model development; conclusions are bounded by the scope and potential biases of reviewed sources. Second, the generative AI landscape evolves rapidly — new capabilities, detection approaches, and regulatory developments may have emerged between study completion and publication. Third, despite systematic search procedures, the review may not capture all relevant grey literature, industry reports, or non-English publications. Fourth, comparative governance analysis is necessarily high-level; comprehensive comparative legal analysis requires specialized expertise beyond this study’s scope. Fifth, synthesis of a rapidly evolving technical field involves interpretive judgments that may reflect the

authors’ collective domain limitations. These limitations do not invalidate the findings but define the scope within which they should be applied.

10 Discussion and Findings

This section synthesizes the evidence presented in Sections 5 through 8 and provides direct, structured answers to the two research questions stated in Section 3. The discussion moves from findings to interpretation, drawing connections across the technical, ethical, and governance dimensions of the paper.

10.1 RQ1: Ethical Challenges of Deepfake Technologies

RQ1: What ethical challenges arise from the development and use of deepfake technologies, particularly regarding consent, identity protection, and human dignity?

The evidence reviewed across Sections 5, 7, and 8 demonstrates that deepfake technologies generate ethical harms across four interconnected dimensions that together constitute a systemic threat to individual rights and social trust.

The most direct harm is the violation of consent and identity ownership. Non-consensual synthetic media — including fabricated intimate imagery and false statements attributed to real individuals — violates personal autonomy from the moment of creation, not merely at distribution. Critically, this breach begins at the model development stage: training data is frequently scraped without the consent of individuals depicted, meaning ethical responsibility is distributed throughout the entire generation pipeline, not only at its end. Once synthetic content is distributed, reputational and psychological damage — including documented outcomes of acute anxiety, depression, and social withdrawal — is rarely fully reversible.

The second major ethical harm is epistemic: the liar’s dividend. Deepfake proliferation erodes public confidence in authentic evidence, and corrections do not fully undo the persuasive impact of synthetic content already consumed. This effect structurally weakens legal and political accountability mechanisms that depend on the credibility of recorded evidence.

Third, deepfake detection systems introduce fairness harms. Detection models perform inconsistently across demographic groups, with marginalized populations experiencing higher false positive

and false negative rates. Deploying such systems at scale without bias auditing compounds existing inequities.

Fourth, accountability is structurally diffuse. Responsibility is fragmented across developers, platforms, distributors, and end users, and existing legal frameworks are poorly equipped to assign liability.

In answer to RQ1: ethical challenges are profound, multi-dimensional, and span the entire content lifecycle from data collection to deployment and detection.

10.2 RQ2: Cybersecurity Risks and Detection Effectiveness

RQ2: What cybersecurity and fraud risks are introduced by deepfake technologies, and how effective are current detection approaches?

Deepfake-enabled threats are actively exploited across financial systems, authentication infrastructure, democratic processes, and legal institutions.

Voice cloning attacks within business email compromise (BEC) campaigns have produced multi-million-dollar losses. Video-based impersonation directly subverts KYC and multi-factor authentication systems. Political disinformation campaigns using AI-generated media have been documented across multiple election cycles.

Four attack types are rated Critical, all exploiting institutional trust relationships at scale.

Current detection approaches are insufficient. No single method performs well across all evaluation dimensions. Visual detection fails on diffusion models, and multimodal approaches are computationally infeasible at scale. Frequency-domain masking shows promise but lacks real-world validation.

In answer to RQ2: deepfake risks are severe and expanding, while detection remains in a persistent adversarial deficit relative to generative capabilities.

10.3 Cross-Cutting Implications

The deepfake challenge is fundamentally a coordination problem. Detection alone cannot address ethical harm; governance alone cannot keep pace with technological change; and ethical frameworks require legal enforcement mechanisms.

Progress requires coordinated investment across detection research, governance frameworks, ethical development practices, platform accountability,

and public media literacy. No single actor can address the challenge independently.

11 Conclusion

This paper has examined deepfake technologies across three interconnected dimensions — technical detection, cybersecurity risk, and governance — and has answered two guiding research questions through a structured synthesis of the 2021–2026 literature. The core findings are clear and mutually reinforcing.

Detection methods are in a persistent adversarial deficit: advances in generative architecture, particularly the shift from GAN-based to diffusion-based generation, consistently outpace improvements in detection. Frequency-domain masking and multimodal ensemble approaches represent the current frontier but face computational barriers to real-time deployment. Cybersecurity threats enabled by deepfakes are not theoretical — they are actively exploited, with voice cloning, video impersonation, and political disinformation producing documented, large-scale harm. Governance frameworks lag significantly behind the technological reality, remaining fragmented across jurisdictions and unable to address the transnational character of synthetic media production and distribution.

Underlying all three dimensions is a shared ethical challenge: deepfake technologies structurally erode the conditions — consent, authentic evidence, accountable authorship — that responsible digital communication depends on. Restoring those conditions requires not a single intervention but a coordinated, interdisciplinary response. This paper contributes a structured synthesis of the current landscape as a foundation for that effort, with the aim of enabling researchers, practitioners, and policymakers to identify where investment and coordination are most urgently needed.

12 Acknowledgments

We gratefully acknowledge Dr. Zainab Albujaasim for her insightful guidance, constructive feedback, and continuous support throughout the development of this work.

13 Individual Contributions

Walther Del Orbe led the literature search and selection, coordinated the overall paper structure, and authored Sections 1 (Introduction) and 7 (Governance). He also managed version control and

final document compilation, and prepared the governance comparison table.

Nawaz Syed authored Section 5 (Cybersecurity and Fraud Risks), researched documented deepfake attack incidents, constructed the attack typology and risk assessment table, and contributed to detection analysis in Section 6.

Kwaku Dei Amoani authored Sections 4 (Methods) and 6 (Detection Techniques), developed the five-dimension detection comparison framework, constructed comparative tables, and led the analysis of detection literature.

Gwendolyn Vongkasemsiri authored Sections 2 (Related Work), 8 (Ethics Statement), 9 (Limitations), 10 (Discussion and Findings), and 11 (Conclusion). She reviewed governance and legal literature, synthesized findings on fairness and bias in detection systems, and performed final copyediting and academic tone review.

References

- [1] Clark, S., & Lewandowsky, S. (2026). The continued influence of AI-generated deepfake videos despite transparency warnings. *Communications Psychology*.
- [2] Doloriel, C., Ullah, H., Liland, K., Machot, F., & Cheung, N. (2025). Towards sustainable universal deepfake detection with frequency-domain masking. *arXiv:2504.07598*.
- [3] European Parliament. (2024). Regulation (EU) 2024/1689 — The Artificial Intelligence Act. *Official Journal of the European Union*.
- [4] Farid, H. (2025). Mitigating the harms of manipulated media: Confronting deepfakes and digital deception. *PNAS Nexus*.
- [5] Gong, L. Y., & Li, X. J. (2024). A contemporary survey on deepfake detection: datasets, algorithms, and challenges. *Electronics*.
- [6] Guarnera, L., Giudice, O., & Battiato, S. (2023). Level up the deepfake detection: Discriminating GAN and diffusion generated images. *arXiv:2303.00917*.
- [7] Khan, N., Nguyen, T., Bermak, A., & Khalil, I. (2025). Unmasking synthetic realities in generative AI: A comprehensive review of adversarially robust deepfake detection systems. *arXiv:2501.09283*.
- [8] Bateman, J., et al. (2025). Deepfake detection in generative AI: A legal framework proposal to protect human rights. *Computer Law & Security Review*. <https://doi.org/10.1016/j.clsr.2025.01.002>
- [9] Lin, G., Lin, L., Walker, C., Schiff, D., & Hu, S. (2025). Fit for purpose? Deepfake detection in the real world. *arXiv:2503.11143*.
- [10] Liu, B., Zhu, T., & Ding, M. (2025). A review of deepfake and its detection: From generative adversarial networks to diffusion models. *International Journal of Intelligent Systems*.
- [11] Martinek, A. (2025). Detecting deepfakes and false ads through analysis of text-based transcripts. In *Proceedings of COLING 2025*.
- [12] Patel, R., et al. (2026). AI-driven conceptual framework for detecting fake news and deepfake content. *Digital Health & Media*, 4(1).
- [13] Singh, S., Srivastav, S., Singh, S., & Hussain, S. (2025). Deepfake detection systems: A comprehensive survey of algorithms and techniques. *Journal of AI Research*.
- [14] Soundarya, B. C., & Gururaj, H. L. (2026). Deepfake detection: Critical review of state-of-the-art approaches and future perspectives. *Discover Applied Sciences*.
- [15] Soundarya, B. C., et al. (2025). Multimedia-enabled deepfake detection: State-of-the-art tools and techniques and future directions. *Discover Computing*.